OPERATING SYSTEM (Jackfruit)

Demo with Screenshots

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SETUP

TERM 1
cd OS-Jackfruit

```
sudo insmod monitor.ko
sudo mknod /dev/container_monitor c 237 0
sudo chmod 666 /dev/container_monitor

sudo rm -rf rootfs-alpha rootfs-beta
sudo cp -a rootfs-base rootfs-alpha
sudo cp -a rootfs-base rootfs-beta

sudo cp cpu_hog rootfs-alpha/
sudo cp cpu_hog rootfs-beta/
sudo cp memory_hog rootfs-alpha/
sudo cp memory_hog rootfs-beta/
sudo cp io_pulse rootfs-alpha/

sudo ./engine supervisor
```

3.1 Multi-container Supervision

Shows two containers running under a single supervisor process.

TERM-2

1&2:
sudo chmod 666 /tmp/engine_cmd

echo "start alpha ./rootfs-alpha /cpu_hog" > /tmp/engine_cmd
echo "start beta ./rootfs-beta /cpu_hog" > /tmp/engine_cmd
echo "ps" > /tmp/engine_cmd

```
[Supervisor] Started alpha (PID 3244, nice=0)

===== CONTAINERS =====
ID: alpha | PID: 3244 | Running | Uptime: 4 sec
======================
[Supervisor] Started alpha (PID 3261, nice=0)
[Supervisor] Started beta (PID 3265, nice=0)

===== CONTAINERS =====
ID: alpha | PID: 3244 | Running | Uptime: 111 sec
ID: alpha | PID: 3261 | Running | Uptime: 0 sec
ID: beta | PID: 3265 | Running | Uptime: 0 sec
```

3.2 Metadata Tracking

Displays container metadata using `engine ps`.

```
===== CONTAINERS =====
ID: alpha | PID: 3244 | Running | Uptime: 111 sec
ID: alpha | PID: 3261 | Running | Uptime: 0 sec
ID: beta | PID: 3265 | Running | Uptime: 0 sec
======================
```

3.3 Bounded-buffer Logging

Shows logs captured through producer-consumer pipeline.

```
cp -a rootfs-base rootfs-logger
sudo cp io_pulse rootfs-logger/
echo "start logger ./rootfs-logger /io_pulse" > /tmp/engine_cmd
sleep 5
```
cat logger.log

```
[04/21/26]seed@VM:~/OS-Jackfruit$ sudo cp -a rootfs-base rootfs-logger
[04/21/26]seed@VM:~/OS-Jackfruit$ sudo cp io_pulse rootfs-logger/
[04/21/26]seed@VM:~/OS-Jackfruit$ echo "start logger ./rootfs-logger /io_pulse" > /tmp/engine_cmd
[04/21/26]seed@VM:~/OS-Jackfruit$ sleep 5
[04/21/26]seed@VM:~/OS-Jackfruit$ cat logger.log
io_pulse wrote iteration=3
io_pulse wrote iteration=7
io_pulse wrote iteration=11
io_pulse wrote iteration=15
io_pulse wrote iteration=19
[04/21/26]seed@VM:~/OS-Jackfruit$
```

3.4 CLI and IPC

Demonstrates CLI communicating with supervisor via UNIX socket.

sudo cp -a rootfs-base rootfs-delta
sudo cp cpu_hog rootfs-delta/
echo "start delta ./rootfs-delta /cpu_hog" > /tmp/engine_cmd
echo "ps" > /tmp/engine_cmd
echo "stop delta" > /tmp/engine_cmd
echo "ps" > /tmp/engine_cmd

```
[Supervisor] Started delta (PID 3321, nice=0)

===== CONTAINERS =====
ID: alpha | PID: 3244 | Running | Uptime: 1092 sec
ID: alpha | PID: 3261 | Running | Uptime: 981 sec
ID: beta | PID: 3265 | Running | Uptime: 981 sec
ID: logger | PID: 3293 | Running | Uptime: 561 sec
ID: delta | PID: 3321 | Running | Uptime: 15 sec
======================
[Supervisor] Stopped delta

===== CONTAINERS =====
ID: alpha | PID: 3244 | Running | Uptime: 1123 sec
ID: alpha | PID: 3261 | Running | Uptime: 1012 sec
ID: beta | PID: 3265 | Running | Uptime: 1012 sec
ID: logger | PID: 3293 | Running | Uptime: 592 sec
ID: delta | PID: 3321 | Stopped | Uptime: 46 sec
======================
```

3.5 Soft-limit Warning

sudo dmesg -C
sudo `cp -a rootfs-base rootfs-mem`
`sudo cp memory_hog rootfs-mem/`
`echo "start memtest ./rootfs-mem /memory_hog" > /tmp/engine_cmd`
`TERM3`
`5&6:`
`sudo dmesg -w | grep -i monitor`

Kernel logs showing memory soft limit exceeded.

```
[04/21/26]seed@VM:~/OS-Jackfruit$ sudo dmesg -w | grep -i monitor
[ 6080.899292] [monitor] Registered PID 3357 (soft=40000, hard=60000)
[ 6081.539807] [monitor] PID 3357 RSS: 8764 KB
[ 6084.609988] [monitor] PID 3357 RSS: 33372 KB
[ 6087.682491] [monitor] PID 3357 RSS: 57924 KB
[ 6087.683290] [monitor] SOFT limit exceeded PID 3357
```

3.6 Hard-limit Enforcement

Kernel kills container when hard limit is exceeded.

```
[04/21/26]seed@VM:~/OS-Jackfruit$ sudo dmesg -w | grep -i monitor
[ 6080.899292] [monitor] Registered PID 3357 (soft=40000, hard=60000)
[ 6081.539807] [monitor] PID 3357 RSS: 8764 KB
[ 6084.609988] [monitor] PID 3357 RSS: 33372 KB
[ 6087.682491] [monitor] PID 3357 RSS: 57924 KB
[ 6087.683290] [monitor] SOFT limit exceeded PID 3357
[ 6090.753484] [monitor] PID 3357 RSS: 82476 KB
[ 6090.753696] [monitor] HARD kill PID 3357
```

```
----------------------
[Supervisor] Started memtest (PID 3121, nice=0)

===== CONTAINERS =====
ID: alpha | PID: 3034 | Running | Uptime: 762 sec
ID: beta | PID: 3041 | Running | Uptime: 708 sec
ID: logger | PID: 3064 | Running | Uptime: 606 sec
ID: delta | PID: 3090 | Stopped | Uptime: 452 sec
ID: memtest | PID: 3121 | Running | Uptime: 133 sec
======================
```

3.7 Scheduling Experiment

yes > /dev/null &
yes > /dev/null &
sleep 1

p=$(pgrep yes | head -n2)

sudo renice -10 -p <pid1>
sudo renice 10 -p <pid2>

watch -n 1 "ps -eo pid,ni,pcpu,comm | grep yes"
Comparison of containers with different nice values.

```
Every 1.0s: ps -eo pid,ni,pcpu,comm | grep yes

   4046 -10  100 yes
   4047  10 55.9 yes
```

3.8 Clean Teardown

Shows no zombie processes after shutdown.

```
8:
ps -o pid,ppid,stat,cmd | grep Z
```

```
[04/21/26]seed@VM:~/OS-Jackfruit$ ps -o pid,ppid,stat,cmd | grep Z
   5372    5334 S+   grep --color=auto Z
```